Online Donation System - Full Implementation
AGILE DEVELOPMENT PROCESS AND DEVOPS LAB

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1.App.java

```
package com.example;

public class App {

    public boolean processPayment(double amount, String cardNumber) {
        if (amount <= 0) {
            return false;
        }

        if (cardNumber == null || cardNumber.length() != 16) {
            return false;
        }

        return true;
    }

    public static void main(String[] args) {
        App donation = new App();

        double amount = 1000;
        String cardNumber = "1234567812345678";

        boolean result = donation.processPayment(amount, cardNumber);

        if (result) {
            System.out.println("Donation Successful");
        } else {
            System.out.println("Donation Failed");
        }
    }
}
```

2.Apptest.java

```
package com.example;

import org.junit.Test;
import static org.junit.Assert.*;

public class AppTest {

    @Test
    public void testValidDonation() {
        App app = new App();
        assertTrue(app.processPayment(1000, "1234567812345678"));
    }
```

```
    @Test
    public void testInvalidAmount() {
        App app = new App();
        assertFalse(app.processPayment(-500, "1234567812345678"));
    }

    @Test
    public void testInvalidCard() {
        App app = new App();
        assertFalse(app.processPayment(1000, "1234"));
    }
}
```

3.pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
         http://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.example</groupId>
    <artifactId>donation-app</artifactId>
    <version>1.0-SNAPSHOT</version>

    <dependencies>
        <dependency>
            <groupId>junit</groupId>
            <artifactId>junit</artifactId>
            <version>4.13.2</version>
            <scope>test</scope>
        </dependency>
    </dependencies>

</project>
```

4.jenkinsfile

```
pipeline {
    agent any

    tools {
        maven 'M3'
    }

    stages {

        stage('Checkout Code') {
            steps {
                git branch: 'main',
                url: 'https://github.com/your-repo/online-donation-system.git'
            }
        }
```

```
        stage('Build & Test') {
            steps {
                sh 'mvn clean test'
            }
        }

        stage('Package') {
            steps {
                sh 'mvn package'
            }
        }
    }
}
```

5.DockerFile

```
FROM eclipse-temurin:17-jdk

WORKDIR /app

COPY target/donation-app-1.0-SNAPSHOT.jar app.jar

CMD ["java", "-jar", "app.jar"]
```

6.deployment.yaml

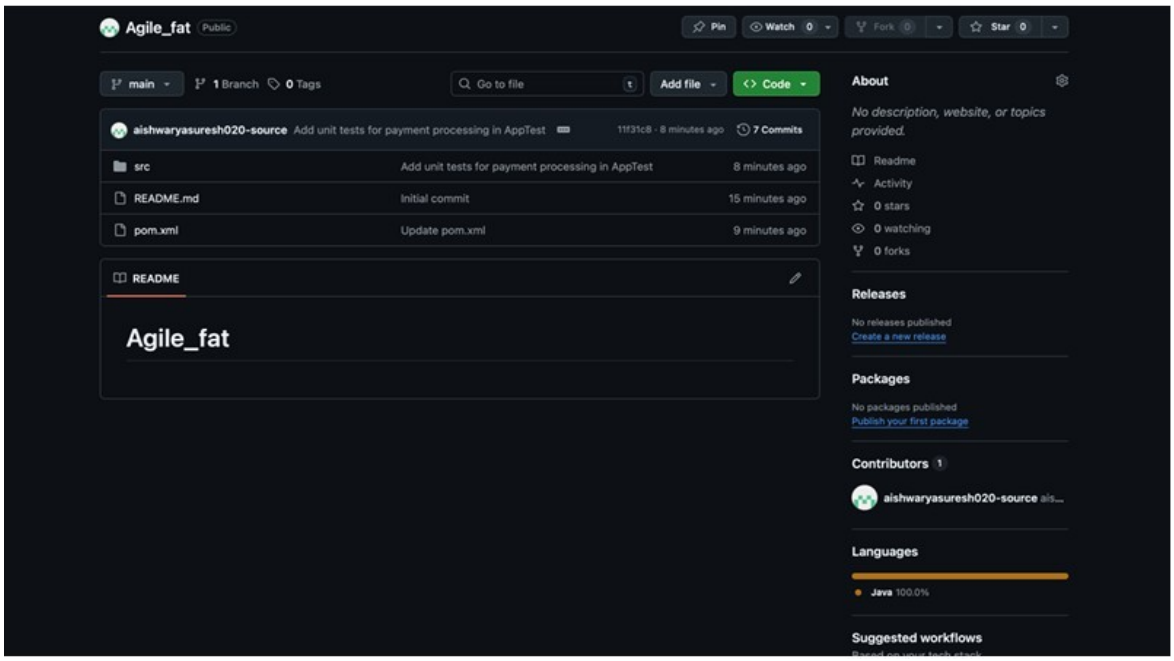
```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: donation-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: donation
  template:
    metadata:
      labels:
        app: donation
    spec:
      containers:
        - name: donation-container
          image: yourdockerhubusername/donation-app:latest
          ports:
            - containerPort: 8080
```

7.Service.yaml

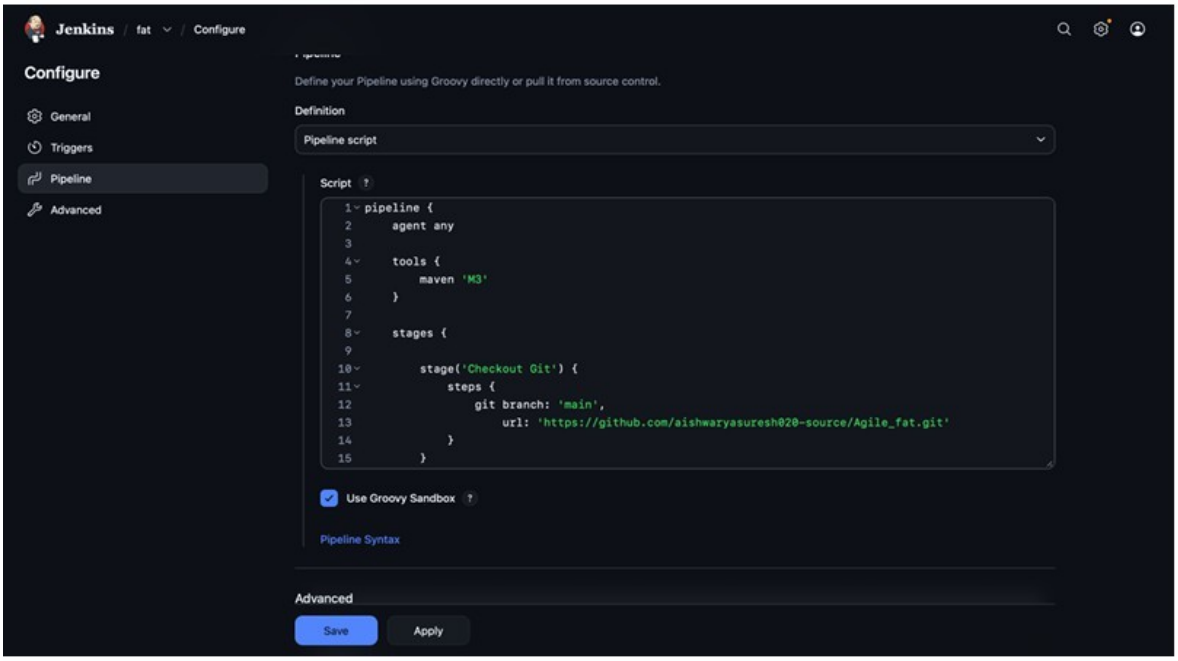
```
apiVersion: v1
kind: Service
metadata:
  name: donation-service
spec:
  type: NodePort
```

```
  selector:
    app: donation
  ports:
    - port: 80
      targetPort: 8080
nodePort: 30099
```

1. GitHub Repository Setup



2. Jenkins Pipeline Configuration



3. Jenkins Build Execution

```
Started by user Aishwarya Suresh
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /Users/aishwaryasuresh/.jenkins/workspace/fat
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Tool Install)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Checkout Git)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
 > git rev-parse --resolve-git-dir /Users/aishwaryasuresh/.jenkins/workspace/fat/.git # timeout=10
Fetching changes from the remote Git repository
 > git config remote.origin.url https://github.com/aishwaryasuresh020-source/Agile_fat.git # timeout=10
Fetching upstream changes from https://github.com/aishwaryasuresh020-source/Agile_fat.git
 > git --version # timeout=10
 > git --version # 'git version 2.39.5 (Apple Git-154)'
 > git fetch --tags --force --progress -- https://github.com/aishwaryasuresh020-source/Agile_fat.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
 > git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 11f31c808136d6f50cdfd28edc76756e4aada624 (refs/remotes/origin/main)
```

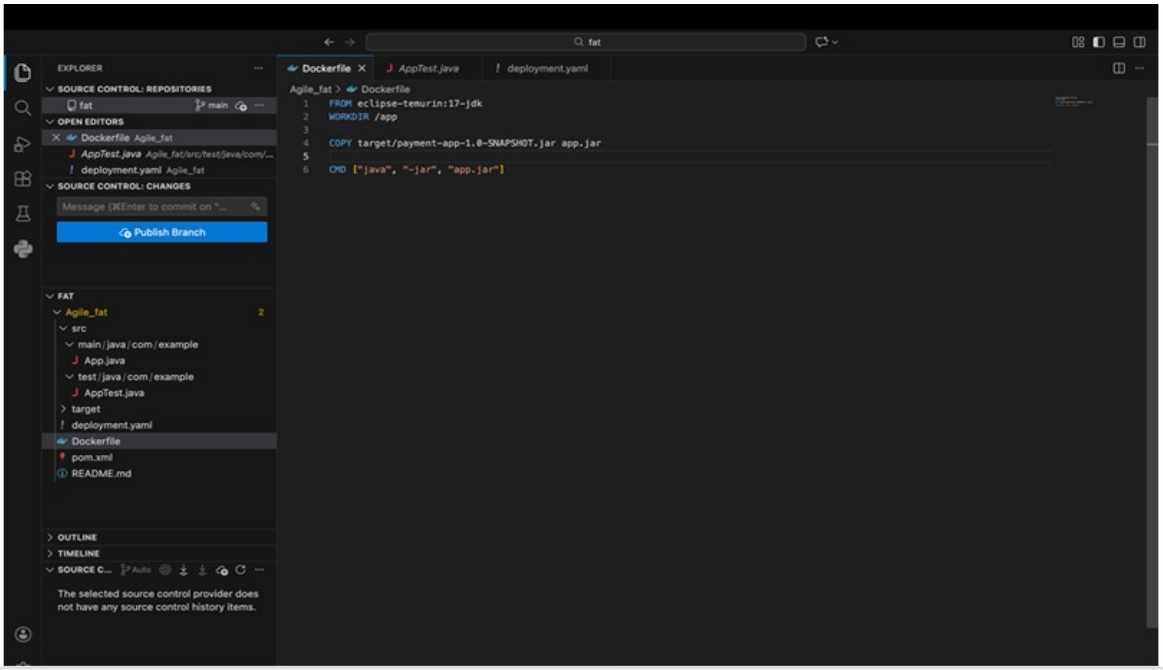
4. Maven Build & Test Output

```
 > git config core.sparsecheckout # timeout=10
 > git checkout -f 11f31c808136d6f50cdfd28edc76756e4aada624 # timeout=10
 > git branch -a -v --no-abbrev # timeout=10
 > git branch -D main # timeout=10
 > git checkout -b main 11f31c808136d6f50cdfd28edc76756e4aada624 # timeout=10
Commit message: "Add unit tests for payment processing in AppTest"
 > git rev-list --no-walk 11f31c808136d6f50cdfd28edc76756e4aada624 # timeout=10
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build and Test)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] sh
+ mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< com.example:payment-app >-----------------------
[INFO] Building payment-app 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ payment-app ---
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ payment-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /Users/aishwaryasuresh/.jenkins/workspace/fat/src/main/resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ payment-app ---
```

5. JUnit Test Results

```
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ payment-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.017 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.808 s
[INFO] Finished at: 2026-04-13T10:25:56+05:30
[INFO] ------------------------------------------------------------------------
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

6. Dockerfile Implementation



7. Docker Build & Push

```
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Agile_fat % mvn clean package
[INFO] -------------------------------[ jar ]-------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ payment-app ---
[INFO] Deleting /Users/aishwaryasuresh/Downloads/fat/Agile_fat/target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ payment-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /Users/aishwaryasuresh/Downloads/fat/Agile_fat/src/main/resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ payment-app ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target/classes
[WARNING] bootstrap class path is not set in conjunction with -source 8
  not setting the bootstrap class path may lead to class files that cannot run on JDK 8
    --release 8 is recommended instead of -source 8 -target 1.8 because it sets the bootstrap class path automatically
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ payment-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /Users/aishwaryasuresh/Downloads/fat/Agile_fat/src/test/resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ payment-app ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target/test-classes
[WARNING] bootstrap class path is not set in conjunction with -source 8
  not setting the bootstrap class path may lead to class files that cannot run on JDK 8
    --release 8 is recommended instead of -source 8 -target 1.8 because it sets the bootstrap class path automatically
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ payment-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.017 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ payment-app ---
[INFO] Building jar: /Users/aishwaryasuresh/Downloads/fat/Agile_fat/target/payment-app-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.799 s
[INFO] Finished at: 2026-04-13T10:29:56+05:30
[INFO] ------------------------------------------------------------------------
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Agile_fat %
```

8. Kubernetes Deployment

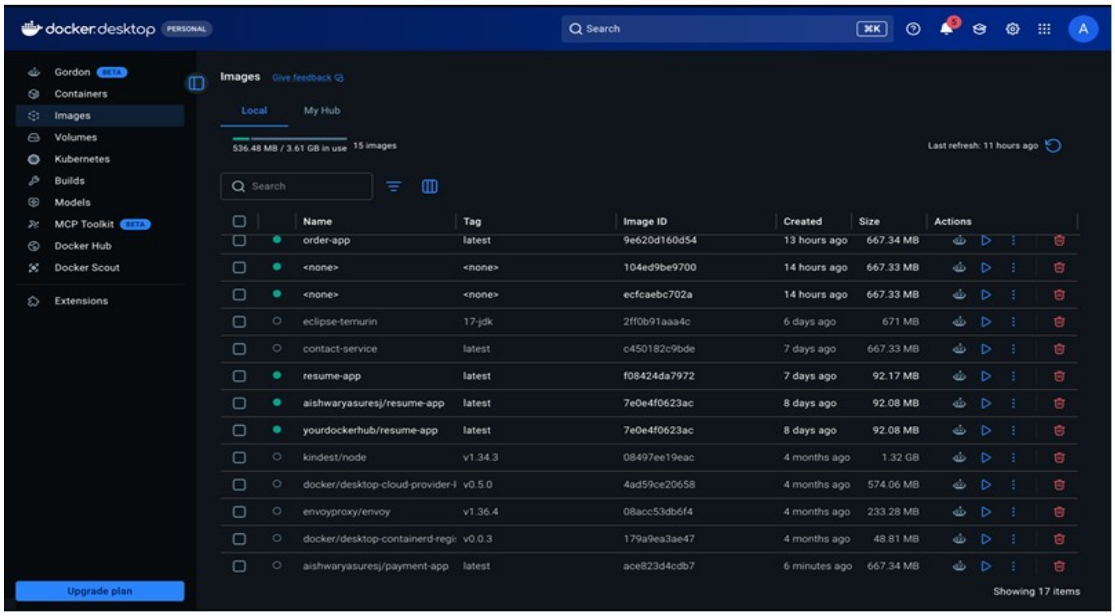
```
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Agile_fat % docker build -t aishwaryasuresj/payment-app .
[+] Building 0.1s (8/8) FINISHED                                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                                0.0s
 => => transferring dockerfile: 160B                                                                                                0.0s
 => [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk                                                          0.0s
 => [internal] load .dockerignore                                                                                                   0.0s
 => => transferring context: 2B                                                                                                     0.0s
 => [1/3] FROM docker.io/library/eclipse-temurin:17-jdk@sha256:2ff0b91aaa4cb5dc35b4760c37fe064390a486dcd90b361caa72b55d598601f3      0.0s
 => => resolve docker.io/library/eclipse-temurin:17-jdk@sha256:2ff0b91aaa4cb5dc35b4760c37fe064390a486dcd90b361caa72b55d598601f3      0.0s
 => [internal] load build context                                                                                                   0.0s
 => => transferring context: 2.44kB                                                                                                 0.0s
 => CACHED [2/3] WORKDIR /app                                                                                                       0.0s
 => [3/3] COPY target/payment-app-1.0-SNAPSHOT.jar app.jar                                                                          0.0s
 => exporting to image                                                                                                              0.0s
 => => exporting layers                                                                                                             0.0s
 => => exporting manifest sha256:2facc22ddec9a7e3eb9cfe0ba8d652f243e6c5af4586ee331ae6d5d1078369e9                                   0.0s
 => => exporting config sha256:617fa76d1b4aeeff52c1ee290f5f33ef8c8186749f27943b182f2ecd3a74c6ed                                    0.0s
 => => exporting attestation manifest sha256:efa4572133dc36550708497 9f193042bf9639f254d75a717f6280 12c29fdfa5b                    0.0s
 => => exporting manifest list sha256:ace823d4cdb789c110e538c784c80c7627f8e6c49eaab4896b08c5d75d6b22e8                              0.0s
 => => naming to docker.io/aishwaryasuresj/payment-app:latest                                                                       0.0s
 => => unpacking to docker.io/aishwaryasuresj/payment-app:latest                                                                    0.0s
```

```
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Agile_fat % docker login
Authenticating with existing credentials... [Username: aishwaryasuresj]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Agile_fat % docker push aishwaryasuresj/payment-app
Using default tag: latest
The push refers to repository [docker.io/aishwaryasuresj/payment-app]
4e985483d507: Mounted from aishwaryasuresj/shopping-cart-app
af0cae6e26c9: Mounted from aishwaryasuresj/shopping-cart-app
6dee3620fe4a: Pushed
36e490e6a042: Pushed
76fd055477b6: Mounted from aishwaryasuresj/shopping-cart-app
62c8686cc493: Mounted from aishwaryasuresj/shopping-cart-app
ec83fb152f23: Mounted from aishwaryasuresj/shopping-cart-app
59b4fba6561b: Mounted from aishwaryasuresj/shopping-cart-app
```

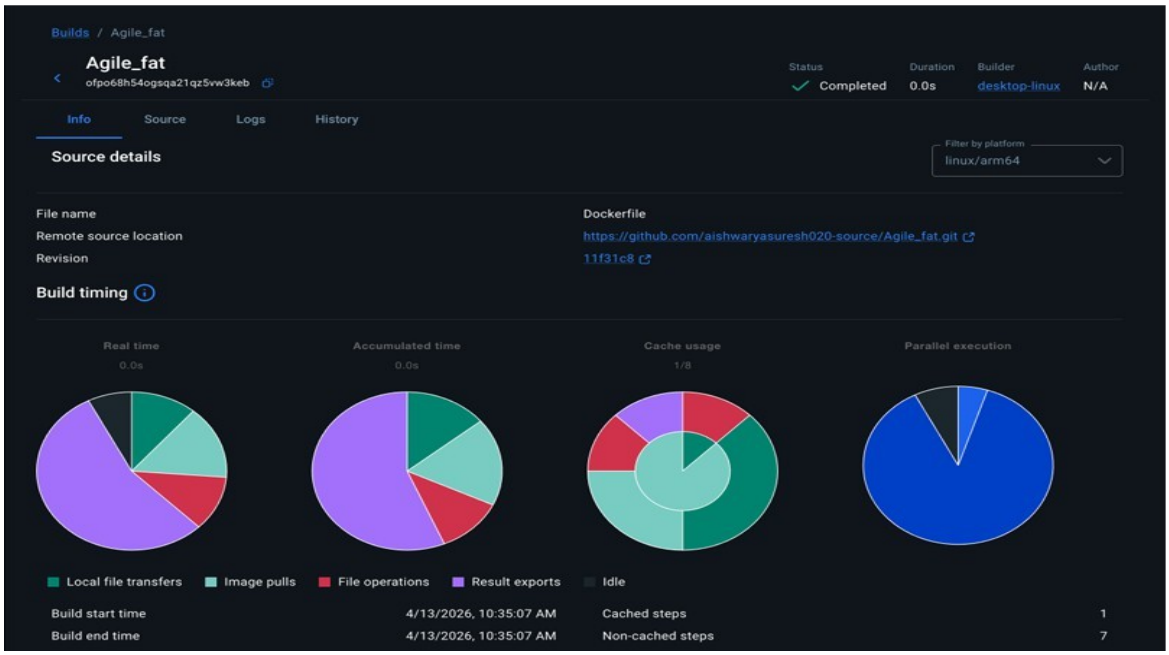
9. Docker Desktop Images

```
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Agile_fat % kubectl apply -f deployment.yaml

deployment.apps/payment-app created
service/payment-service created
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Agile_fat % kubectl get pods
NAME                                READY   STATUS             RESTARTS        AGE
order-app-deployment-74bdc75d97-zf97d   0/1     ErrImageNeverPull   0               12h
payment-app-6b9978f854-5xqks        0/1     ErrImagePull        0               16s
payment-app-6b9978f854-r456d        0/1     ErrImagePull        0               16s
refund-app-8fb9cd9db-f6b7n          0/1     ImagePullBackOff    0               6h29m
refund-app-8fb9cd9db-qj25f          0/1     ImagePullBackOff    0               6h29m
resume-deployment-84c4cc5dfc-kqzcz  1/1     Running             11 (8m48s ago)   7d17h
resume-deployment-84c4cc5dfc-xqz9r  1/1     Running             11 (8m48s ago)   7d17h
shopping-cart-app-75b4d8c96b-5lcrn  0/1     CrashLoopBackOff    31 (2m51s ago)   8h
shopping-cart-app-75b4d8c96b-vv4t7  0/1     CrashLoopBackOff    31 (3m ago)      8h
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Agile_fat % kubectl get svc
NAME              TYPE       CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
kubernetes        ClusterIP  10.96.0.1       <none>        443/TCP        8d
payment-service   NodePort   10.96.201.188   <none>        80:30099/TCP   22s
refund-service    NodePort   10.96.59.111    <none>        80:30003/TCP   6h29m
resume-service    NodePort   10.96.58.224    <none>        80:30007/TCP   7d17h
shopping-cart-app NodePort   10.96.171.189   <none>        80:30009/TCP   8h
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Agile_fat %
```

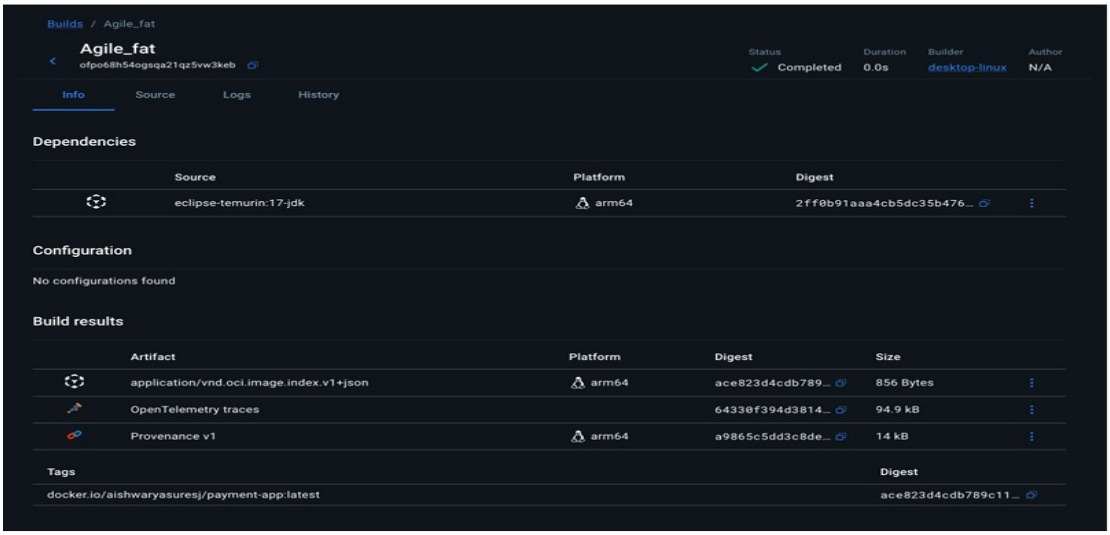
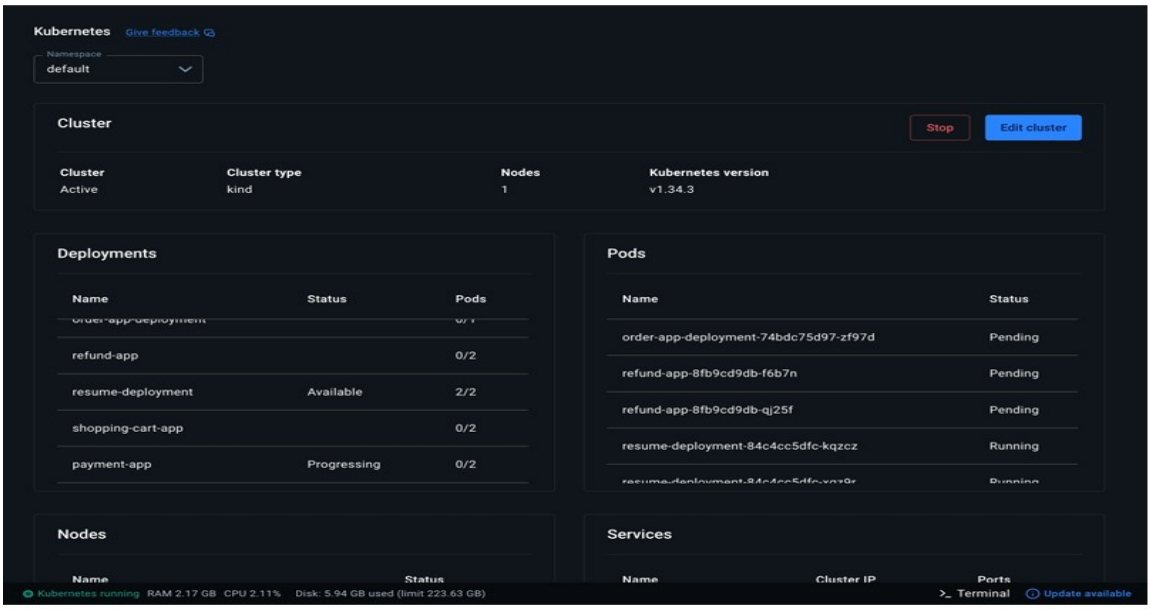
10. Kubernetes Cluster View



11. Push to GitHub



12. Updated GitHub Structure



13. Project Folder Structure